

Preprocessing, Model Comparison, and Model Selection

1. Preprocessing Choices

The Adult Income dataset contains both numerical and categorical features. To prepare the data for machine learning, numerical and categorical features were handled separately using a preprocessing pipeline. Numerical features were processed appropriately for the models, while categorical features were converted into numerical form using one-hot encoding. Using a **ColumnTransformer** and **Pipeline** ensured that the same preprocessing steps were applied consistently to both the training and testing data and helped prevent data leakage.

The preprocessing pipeline was also kept separate from the model so that it can be reused in later experiments. The transformed feature names were obtained using **get_feature_names_out()**, which allowed the Logistic Regression coefficients to be mapped back to meaningful feature names.

2. Model Comparison

Two baseline classification models were evaluated: Logistic Regression and Decision Tree. Logistic Regression was selected because it provides a relatively simple and interpretable model. Its coefficients show which features are positively or negatively associated with predicting income above \$50K. The strongest positive coefficient was **capital-gain** (2.222), while **occupation_Priv-house-serv** had the strongest negative coefficient (-1.411). Other important features included marital status, education, occupation, age-related information, and relationship categories.

The Decision Tree was used to capture nonlinear relationships between the features and the target. It achieved a test accuracy of **81.83%**, with **61.92% precision**, **62.53% recall**, **62.23% F1-score**, **75.22% ROC-AUC**, and **47.69% PR-AUC**. The tree's early splits were based on sensible variables such as marital status, capital gain, education, and age.

3. Model Limitations and Overfitting

Although the Decision Tree achieved reasonable test performance, it showed strong evidence of overfitting. The tree had a depth of **56**, with a training accuracy of **99.99%**, compared with a test accuracy of **81.83%**. This produced a train-test gap of approximately **18.16 percentage points**. This large difference indicates that the unrestricted tree has learned the training data too closely and does not generalize equally well to unseen data.

The confusion matrices also showed that errors should be considered when selecting the model. Logistic Regression produced more false negatives than false positives, while the Decision Tree had relatively similar numbers of false positives and false negatives. Therefore, accuracy alone is

not sufficient for model selection. Precision, recall, F1-score, ROC-AUC, and PR-AUC should also be considered.

4. Model Selection

For further development, **both Logistic Regression and Decision Tree will be continued**, but for different reasons. Logistic Regression will be retained as the main interpretable baseline because its coefficients make the model's predictions easier to understand. The Decision Tree will be further developed because it can capture nonlinear relationships that Logistic Regression may not capture, but the current tree requires tuning before it can be considered a strong final model.

The Decision Tree will be tuned to reduce overfitting by testing parameters such as **max_depth**, **min_samples_split**, and **min_samples_leaf**. Class weighting may also be tested to determine whether the model can improve its handling of the positive income class. Future models will be compared using F1-score, recall, ROC-AUC, and PR-AUC in addition to accuracy.

5. Conclusion

Overall, the preprocessing approach provided a consistent and reusable foundation for both models. Logistic Regression offers better interpretability, while the Decision Tree provides a way to model nonlinear relationships. The current Decision Tree is clearly overfitting, so it will require tuning before further conclusions are made. Therefore, both models will be carried forward to the next stage, with Logistic Regression serving as the interpretable baseline and a regularized/tuned Decision Tree being tested as the nonlinear alternative.